Enabling Computer Education through Gaming Factors : A Cas e S tudy on Improving Learning Efficiency

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Summary

Gamification has gained attention as a technique that positively impacts on increasing motivation and learning outcomes for learners across various educational fields. This study aims to analyze whether such positive effects of gamification can also be applied in the field of computer education and what those effects might be. The study was conducted by tracking changes in the academic performance trends of a student group where gamification techniques were applied (hereafter Group A) and a student group where they were not applied (hereafter Group B). The analysis revealed that Group A showed a significant improvement in academic performance compared to Group B, and a difference in final grades from the control group could be confirmed. This result demonstrates that gamification can enhance achievement, promote immersion and engagement, similar to its context in other educational fields.

1. Introduction

The development of digital technology has brought a revolution in traditional education. Among these, gamification is being utilized as a tool to motivate learners and maximize learning effectiveness. Gamification refers to the use of game design elements in non-game environments [1]. Recent studies show that gamification can enhance learning motivation and performance, but researches on the specific positive effects this technique has on computer education remain limited. This paper aims to analyze the positive impact of gamification techniques on computer education.

2. Theoretical Background

2.1 Universal Effects of Gamification

According to existing researches, gamification has a positive effect on improving learners' academic abilities, regardless of region, educational level, learning environment, subject matter, or the specific game elements used[2]. This universal effect suggests that gamification is not limited to specific subjects or learning environments but can yield similar positive outcomes across diverse educational contexts. As demonstrated by Zeng, J.'s research, gamification is adaptable as a tool for enhancing learning outcomes in various situations. This study aims to demonstrate by applying these universal effects to computer education that gamification can make a substantial contribution to enhancing learning motivation and performance.

2 . 2 Applying Gamification in Computer Education

Computer education, by its very nature, demands logical thinking and problem-solving skills, and learners may struggle with these complex concepts. Gamification techniques are effective at increasing learners' immersion and helping them understand complex concepts more intuitively, so it is believed they could yield similar benefits in computer education. Previous researches have shown that introducing gamification into education improves logical thinking and problem-solving skills[3], supporting the hypothesis that applying gamification to computer education will also yield positive effects.

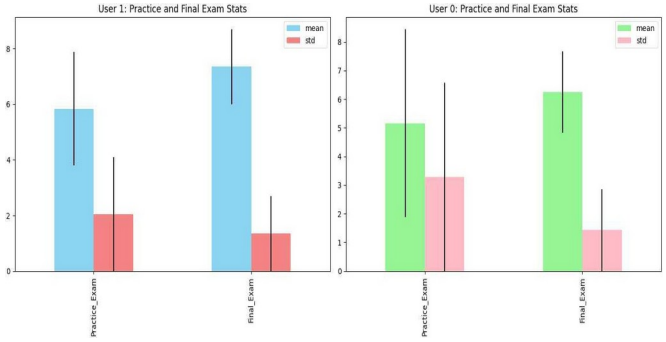
3. Main Body of Paper

As mentioned earlier, gamification has been proven to be a powerful tool for enhancing learning outcomes across various educational fields. Based on this, gamification is also highly likely to positively influence students' immersion and achievement in computer education. Based on this theoretical background, this study aims to empirically analyze how gamification affects learning outcomes in computer education.

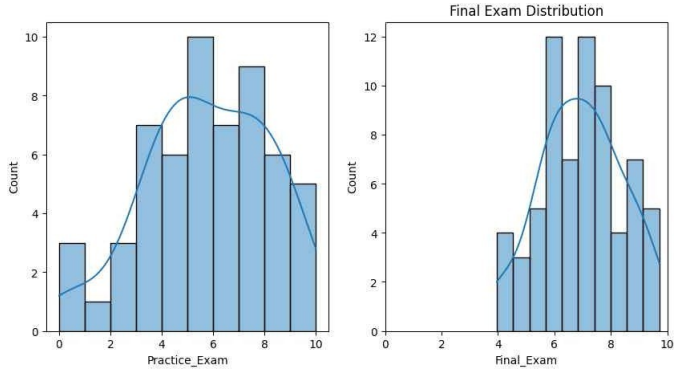
This study addresses the following primary research question: 'How does gamification affect learners' performance improvement in computer education?' Accordingly, the hypothesis is: 'The group with gamification applied will show significantly improved learning performance compared to the control group.'

This study was conducted with a sample of 69 students enrolled in a computer science course. Participants were divided into an experimental group that had gamification applied and a control group that received only traditional education methods. The experimental group performed learning activities utilizing game-based elements (flashcards, badges, etc.) during class, and their performance changes were analyzed by measuring scores on the initial test, the midterm exam in week 6, and the final exam. Data analysis used t-tests to verify the difference in performance between the two groups.

Figure 1 visually represents the results of the first and final tests between the control group and the gamification group. This shows that both gamified education and traditional education contributed to academic improvement, but the increase from gamified education was significantly steeper. Additionally, Figure 2 shows the distribution of student performance scores throughout the study.

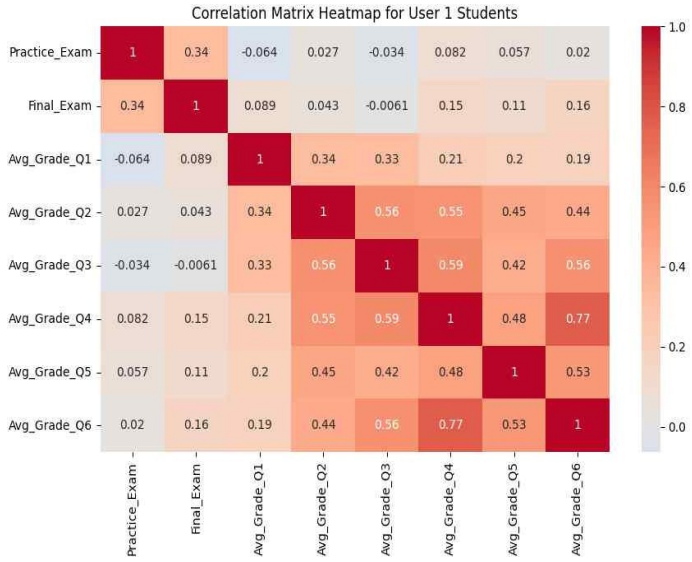


(Figure 1)



(Figure 2)

Figure 3 shows the results of expressing the 6-week academic performance trend of the gamification group as a correlation matrix. As seen by the fact that the colors become darker the closer they are to the bottom right, i.e., as the education sessions progress, it can be confirmed that gamified education and academic improvement have a high correlation.



(Figure 3)

For data verification, a t-test was conducted by grouping the gamification group and the control group. In the initial test, the difference in scores between the two groups was not significant (p=0.33). However, as the education progressed, the score gap widened, and a significant difference in scores was observed in the final test (p=0.08).

The improvement in performance observed in the gamification group during the midterm and final exams demonstrates that gamification is effective in sustaining learning motivation and immersion. This confirms that gamification aids in enhancing the ability to understand complex concepts and solve problems in computer education. However, as this study targeted a small number of specific classes, there is a need to secure data from diverse groups to obtain more accurate data. Furthermore, examining the standard deviation of each group reveals that the initial scores in the gamified education group were slightly higher, leaving room for regret that variable control should have been managed more strictly.

This study demonstrates that gamification is effective in enhancing learning outcomes in computer education. These results provide a basis for the broader application of gamification techniques in future computer science education. In further researches, it is necessary to verify the long-term effects of gamification across more diverse learning environments and learner groups.

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